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IPSG GUIDELINES & THEIR ROLE IN ENHANCING PATIENT SAFETY & REDUCING CLINICAL ERRORS

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ABSTRACT:

Patient safety is a fundamental component of healthcare quality, with preventable clinical errors remaining a major concern. This study evaluates compliance with International Patient Safety Goals (IPSG) and their role in reducing clinical errors in a tertiary care hospital in India. A descriptive cross-sectional audit-based study was conducted using structured checklists and retrospective medical record reviews, covering 2,524 IPSG audits and 1,177 medical record audits. The findings showed high compliance with patient identification, medication safety, and infection prevention, while gaps were observed in communication, surgical safety, and fall prevention documentation. The study also identified differences between documented practices and real-time clinical practices. It concludes that effective IPSG implementation can strengthen patient safety, supported by continuous staff training, regular audits, leadership commitment, and a strong patient safety culture.

Keywords: *Patient Safety, International Patient Safety Goals, Clinical Errors, Healthcare Quality, Hospital Audit.*

INTRODUCTION

Putting patient safety first sits at the core of what makes healthcare work well. Built into every system, it shapes how reliably care gets delivered. When treatment happens, avoiding injury becomes a top priority. At the same time, reducing avoidable dangers tied to medical decisions matters just as much. This part of care does not draw attention - yet it holds everything together.

Nowadays, as medicine moves faster than ever, keeping patients safe isn't just about skill - it reflects how well organizations manage care. Despite big leaps in research and hospital design, mistakes that could have been prevented still pop up across clinics worldwide. Often hidden, these slipups affect recovery rates while quietly weakening trust in health services. What shifted global attention to patient safety was the release of the Institute of Medicine's "To Err is Human" in 2000. This document revealed how countless deaths occurred annually from avoidable mistakes in care - contrary to earlier views that such problems stemmed mainly from individual failure. Instead, evidence pointed at deeper flaws rooted in how health systems operated. As a result, discussions evolved; studies into safer care began focusing less on personal blame and more on workflows, institutional designs, and broader operational patterns behind errors.

Throughout a patient's care path, mistakes may occur at various stages and at various steps. Skipping these steps led to problems - diagnosis delayed, meds missed, operations postponed. Treatment plans lacked clarity, talks between staff grew unclear. Records stayed incomplete, checks afterward never happened, up after treatment. Mistakes sometimes emerge when health workers fail to share information clearly, while inconsistent documentation adds further risk. Working long hours often leads to mistakes. Following guidelines becomes harder when routines are ignored. Fatigue mixes with understaffing to weaken outcomes. Training shortages show up clearly during emergencies. Problems exist within how things operate. Fatal outcomes may arise, ranging through minor issues up to permanent impairment. Seriousness increases gradually, yet unpredictably, ending sometimes in loss of life.

What happens when mistakes occur in medicine? Costs go up. Hospital stays grow extended because of them. Legal risks rise alongside. Another outcome appears too. Reduced well-being often follows when care misses patient expectations. These days, healthcare setups aren't just busier - they've grown trickier to manage, which often leads to mistakes. Complexity creeps in slowly, then suddenly things go off track. With so many moving parts, small missteps can spiral fast. The way services connect now increases pressure on workers. When workflows get tangled, even careful people slip up. Mistakes happen not because of laziness - but because structures demand too much at once. A small misunderstanding, perhaps just a missed signal, might set off

reactions nobody expected. One misstep, followed by another, pulls things downhill fast. A message delayed, then ignored - suddenly tension grows. Simple oversights, when left unchecked, shift the whole dynamic. Because worries about patient well-being are rising, global health groups along with official certification bodies have emerged with the organized models designed to bring consistency to patient care routines in medical environments.

The Six Patient Safety Goals by JCI have embedded these approaches across healthcare systems worldwide, where their uptake now exceeds that of earlier models worldwide. This model came into use through IPSG, aiming to guide work toward patient care zones with higher likelihood of impact. Injured individuals need support while developing guidelines rooted in data to lower dangers throughout healthcare settings.

The IPSG framework focuses on six critical areas of patient safety:

1. **Correct patient identification**

   The first goal emphasizes the accurate identification of patients using at least two identifiers to prevent errors such as wrong-patient medication administration, incorrect procedures, or misdiagnosis.

2. **Effective communication**

   The second goal focuses on improving effective communication among healthcare providers to ensure timely and accurate exchange of patient information.

3. **Safety of high-alert medications**

   The third IPSG goal addresses the safe use of high-alert medications, recognizing that certain medications carry a higher risk of causing serious harm if used incorrectly.

4. **Safe surgical practices**

   The fourth goal aims to ensure safe surgery by emphasizing correct patient identification, procedure verification, and site confirmation. Surgical errors, although preventable, can have devastating consequences, making adherence to surgical safety protocols essential.

5. **Prevention of healthcare-associated infections**

   The fifth goal focuses on reducing the risk of healthcare-associated infections through evidence-based infection prevention practices like hand hygiene, sterilization, and appropriate use of personal protective equipment.

6. **Reduction of patient fall risks**

The sixth goal addresses the reduction of patient injury brought on by falls, particularly among vulnerable patient populations like elderly or critically ill.

The use of IPSG has become increasingly important in the Indian healthcare sector. Tertiary care hospitals have specialized services , and advanced technologies but they often deal with high patient loads workforce shortages , operational pressures and variability in how staff are trained.

These kinds of problems can affect the consistency of patient safety efforts and make the risk of clinical errors higher. With accreditation standards like the National Accreditation Board for Hospitals and Healthcare Providers (NABH) , Indian hospitals have been pushed to put in place patient safety frameworks that fit international standards. But even if the paperwork looks good, it doesn't always mean the same thing happens in real clinical practice. Very often there is a split between what is documented as compliance and what is actually followed in real time with safety protocols. This mismatch may come from behavioral factors, organizational culture, communication barriers , or workflow inefficiencies.

So, it becomes important to evaluate how IPSG is complied with in real world clinical settings , to gauge how effective the patient safety systems are and to find where improvements are needed.

Audit-based evaluations give a more direct view of what is really happening, and they also provide healthcare organizations a way to check compliance , locate weak spots, and then take specific steps. This study was undertaken to assess adherence to the International Patient Safety Goals in a tertiary care facility in India and to evaluate their role in enhancing patient safety and reducing clinical errors. The study aims to assess the current state of compliance across all six domains of the IPSG, determine the relative strengths and weaknesses of their implementation, and explore the association between structured safety practices and clinical outcomes. This study provides useful insights for healthcare administrators, quality managers and clinical leaders who wish to improve patient safety systems through an audit-based analysis of IPSG implementation. Finally policy development is a necessary part to improve patient safety, but it alone is not enough. Continuous vigilance, staff engagement, leadership commitment and a culture of learning, and accountability are all required too. Healthcare organizations can move closer to the goal of providing safe, dependable and high-quality patient care by systematically assessing what is happening, and then making targeted improvements.

LITERATURE REVIEW

Patient safety is now becoming one of the most important areas within healthcare research and quality improvement pretty much everywhere in the world. It is basically about avoiding unnecessary harm to patients when they're getting health care, and bringing the treatment risks down to something like as low as reasonably

achievable. Over the last 20 years, healthcare organizations and researchers have been slowly acknowledging that patient safety depends on more than just individual clinical competence , you know. It's also tied to system processes, the organizational culture, and those quality assurance mechanisms. Research consistently demonstrates that structured Safety systems are crucial for minimizing risks and enhancing patient results (Vincent, 2010, https://doi.org/10.1002/9781444323856).

The contemporary patient safety movement gained traction following the publication of the landmark report *To Err is Human*, which highlighted the alarming prevalence of preventable medical mistakes in healthcare systems. This report fundamentally transformed worldwide comprehension of patient safety by emphasizing that clinical errors often result from system failures rather than isolated acts of negligence. It shifted the focus from blaming individual practitioners to identifying organizational vulnerabilities and implementing systemic safeguards. The report established a foundation for structured patient safety interventions and continues to influence healthcare quality improvement efforts worldwide (Kohn, Corrigan, & Donaldson, 2000, https://doi.org/10.17226/9728).

Major conceptual framework underpinning patient safety research is Reason's Swiss Cheese Model, which explains how adverse events occur when multiple system weaknesses align. According to this model, healthcare systems consist of multiple layers of defense, each designed to prevent errors. However, each layer contains vulnerabilities or "holes." When these vulnerabilities align, errors pass through all barriers and result in patient harm. This model highlights the significance of multilayered defenses, continuous monitoring, and system redesign in reducing clinical errors. It provides a strong theoretical basis for the application of organized patient safety frameworks such as the Global Patient Safety Objectives (Reason, 2000, https://doi.org/10.1136/bmj.320.7237.768).

Research on patient safety has consistently identified communication failures as one of the leading contributors to unpleasant outcomes that could be avoided communication breakdowns may occur during patient handovers, interdepartmental transfers, verbal order exchanges, or emergency situations. Ineffective communication often leads to delayed interventions, incomplete information transfer, inappropriate treatment decisions, and avoidable complications. Research on structured handoff interventions has demonstrated significant reductions in medical errors when standardized communication tools are used. A large multicenter study examining handoff improvement programs reported substantial reductions in to unpleasant outcomes that could be avoided after implementation of standardized communication practices. (Starmer et al., 2014, https://doi.org/10.1056/NEJMsa1405556).

Medication safety remains another extensively studied aspect of patient safety because of the high frequency and potential severity of medication-related errors. Medication errors can occur at multiple stages, including prescribing, dispensing, preparation, administration, and monitoring. These errors are particularly dangerous when they involve high-alert medications, which carry a greater risk of causing serious patient harm if used incorrectly. Research examining adverse medication reactions found that many medication-related incidents are preventable through structured safeguards and improved system design. These findings reinforce the need of implementing rigorous medication safety protocols (Bates et al., 1995, https://doi.org/10.1001/jama.274.1.29).

Surgical safety has also attracted a lot of attention in patient safety literature because of the potentially catastrophic consequences of procedural errors. Surgical care involves complex coordination among multidisciplinary teams, making communication, verification, and protocol adherence essential. The presentation of surgical safety checklists has been the most impactful patient safety interventions in modern healthcare. Evidence from multicenter international studies has demonstrated that surgical safety checklists significantly reduce morbidity and mortality rates by ensuring that critical verification steps are completed. When checklist compliance becomes a routine administrative task rather than an active safety practice, its impact diminishes considerably (Haynes et al., 2009, https://doi.org/10.1056/NEJMsa0810119).

The International Patient Safety Goals sort of bring together these evidence based ideas into a general structure , to avoid harm that could be prevented. It’s something Joint Commission International developed, and the IPSG covers six zones of risk like patient identification, communication, medication safety, surgical safety, infection prevention, and fall prevention , all in one place. Most research into IPSG implementation reveals gains - fewer errors, sharper attention among teams, records that reflect real progress. Yet hurdles remain despite these patterns emerging across settings.

RESEARCH GAP

Across the world, attention to patient safety in medical studies has expanded significantly - yet gaps remain. Though awareness rises, consistent improvements lag behind. Progress exists; it is uneven. Research continues, though answers stay incomplete. Efforts multiply, even when results fall short. Focus sharpens over time, still challenges persist. Knowledge builds slowly, without sweeping change. Work advances, just not everywhere at once. Still, major gaps exist in understanding how organized safety systems - such as the International Patient Safety Goals - actually function

Putting IPSG into real-world use often faces hurdles. Looking into why mistakes happen in healthcare, plus how to stop them, has taken up plenty of attention lately - plenty of studies exist. Some suggestions emerged from the work done earlier, yet safety research often focuses on isolated parts rather than examining the entire system together. Now their attention shifts abruptly, making links between ideas harder to follow. Suddenly clarity fades when priorities move away from the original point. What was once connected now feels scattered without warning. What stands out most is how scattered the studies appear, lacking any real effort to connect findings across papers.

Patient safety is not one thing, it is multi-dimensioned, and each IPSG goal is basically a piece of one linked system, so if you only assess separate components you end up limiting the view of institutional compliance, and also the overall effectiveness.

One issue that has been kind of underexplored is the gap between documented compliance and what actually happens in clinical practice, not just on paper. Many hospitals have records that show they adhere to patient safety protocols, but observational studies often end up showing inconsistencies, like how those protocols are really carried out in daily routines. For example, checklists might be filled out in a formal way, yet there can be a lack of consistent active participation, and the protocol might not be implemented in the same manner every time. This gap between paper-based compliance and behavioral compliance still has not been looked at enough, at least not in a careful way, within the existing literature. Another major research void is in the Indian health care context.

This study addresses these gaps by offering a thorough audit-based evaluation of all six IPSG domains is a hospital for tertiary care in India. It kind of blends on the spot observation with a check of medical records, to gauge actual compliance in real time. In addition, it helps find mismatches between what is written and what is happening in practice, and it also brings in context-specific proof.

RESEARCH DESIGN

This study used a descriptive cross-sectional, audit based research design to sort of assess adherence to the International Patient Safety Goals (IPSG) and see how much they mattered for patient safety, and also to lower clinical errors in a multispecialty hospital in India The descriptive style felt most suitable, since the study needed to systematically observe, measure and analyze the already existing patient safety practices, without messing with variables or adding any new interventions. Looking at how well each IPSG domain follows rules today formed the core aim - spotting gaps came along with it, and the objectives unfolded, along with their strengths and where execution fell short.

Although the information came from a single timeframe, this study used a cross-sectional structure due to its snapshot nature offering a glimpse into patient safety methods at just one moment in time. Examining various IPSG metrics reveals safety outcomes across an identical period. While one measure shifts, another may hold steady during that span. Each indicator adds context when viewed alongside its peers over those dates. Temporal alignment allows fair comparison between distinct data points. Despite differences in method, timing links their relevance. Patterns emerge only because every series covers the exact interval.

Audit-based methods shaped the study's structure, relying on firsthand observation during site visits. Checklist-guided evaluations followed each session, ensuring consistent criteria across cases. This approach combined different review techniques, leading to a balanced evaluation of how well patient safety was managed along with compliance. Observing healthcare workers during routines revealed how closely they followed safety rules at the moment care happened. Using this method, observation of actual employee actions became possible, revealing how closely they followed established protocols.

The study was conducted across multiple clinical departments within the tertiary care hospital to ensure broad representation of patient safety practices. This cross-departmental approach enhanced the reliability of findings by capturing variations in workflow, operational challenges, and compliance patterns across different care settings.

The descriptive cross-sectional audit-based design was particularly suitable for this study because it enabled objective measurement of compliance, facilitated practical assessment of real-time practices, and generated evidence-based findings relevant for healthcare quality improvement. By using this research model, the study provides a realistic evaluation of IPSG implementation and offers valuable insights into strengthening patient safety systems within tertiary care healthcare institutions.

DATA ANALYSIS & FINDINGS

The present study analyzed adherence to the International Patient Safety Goals (IPSG) using descriptive statistical methods. The data was collected through **2524 direct observational audits** and **1177 medical record audits** conducted across multiple clinical departments of the tertiary care hospital. The analysis focused on measuring compliance across all six IPSG domains and identifying practical strengths and areas requiring improvement. The findings indicate that while the hospital demonstrated a high level of overall compliance with patient safety standards, certain gaps were observed between documented compliance and actual clinical practice.

IPSG 1: - Compliance with Patient Identification Standards

Patient identification showed the highest compliance among all IPSG domains. The overall compliance rate was recorded at **99.9%**, indicating strong institutional adherence to patient verification protocols.

Observations revealed that patient identification was routinely verified using two identifiers, such as patient name and hospital registration number, before medication administration, diagnostic procedures, and sample collection. Wristband usage was consistent across all observed departments, particularly in inpatient wards and intensive care units.

Medical record audits also reflected near-complete documentation of patient identification processes. The consistency observed in both practice and documentation suggests that patient identification protocols have been successfully integrated into routine clinical workflows.

IPSG 2: - Compliance with Effective Communication Protocols

The second IPSG domain, effective communication, recorded a compliance rate of **92%**. The analysis revealed variation in communication practices across departments. Structured communication during shift handovers was more consistent in critical care units, where staff followed detailed reporting formats. However, communication gaps were observed in general wards during periods of high patient activity. **The most common deficiencies included: -** Incomplete handover information, Lack of read-back confirmation for verbal orders, Delayed communication of abnormal findings, Inconsistent escalation procedures.

Medical record reviews showed that communication-related documentation was often incomplete or lacked sufficient detail.

The findings suggest that while communication systems are formally established, practical execution is influenced by workload pressure, staffing patterns, and varying levels of staff attentiveness. This area requires focused improvement due to its direct relationship with preventable clinical errors.

IPSG 3: -Medication Safety Compliance

Medication safety demonstrated a compliance rate of **99%**, indicating a strong medication management system within the hospital.

Observational audits showed proper segregation and labeling of high-alert medications. Nursing staff consistently followed medication verification protocols prior to administration. Double-check systems were observed for high-risk medications, particularly in intensive care and emergency settings.

Key strengths included: - Standardized medication labeling, Secure storage of high-alert drugs, Consistent dose verification, Accurate prescription documentation.

Minor lapses included occasional delays in updating administration records during peak clinical hours. Medical record audits reflected strong consistency between observed practices and documented compliance.

These findings indicate that medication safety systems are well-established and effectively implemented, significantly reducing the likelihood of medication-related adverse events.

IPSG 4: -Surgical Safety Compliance

Compliance with surgical safety protocols was recorded at **98%**, reflecting strong formal adherence to safety checklists. All observed surgical procedures included completion of the required checklist documentation. Verification of patient identity, procedure type, and surgical site was documented in almost every case.

However, observational findings revealed important variations in actual checklist execution. **Specific observations included: -** Inconsistent active participation by all team members, Rapid checklist completion without verbal engagement, Occasional omission of full time-out discussion.

Although documentation suggested high compliance, direct observation indicated that some checklist processes were completed as procedural formalities rather than active safety discussions.

This discrepancy highlights an important distinction between documentation compliance and meaningful implementation. The effectiveness of surgical safety protocols depends not only on checklist completion but also on genuine team communication and engagement.

IPSG 5: - Infection Prevention Compliance

Infection prevention practices demonstrated compliance ranging between **97% and 99%**, indicating strong adherence to established infection control protocols. Hand hygiene compliance was consistently high across critical care and procedural areas. Alcohol-based hand rubs and handwashing facilities were readily accessible. **Observed strengths included: -** Appropriate use of personal protective equipment, Maintenance of sterile fields, Adherence to aseptic techniques, Regular environmental sanitation practices.

Minor deviations were noted during periods of high patient load. Documentation related to infection prevention was complete and regularly maintained. The findings reflect a strong infection prevention culture supported by regular monitoring and institutional reinforcement.

IPSG 6: - Fall Prevention Compliance

Fall prevention compliance ranged between **90% and 94%**, making it one of the lower-performing IPSG domains. Most patients identified as high risk had visible fall precaution indicators and appropriate preventive interventions in place.

Observed preventive measures included: - Bedside rail implementation, Assisted mobility support, Patient and family education.

However, gaps were identified in: - Timely reassessment of fall risk, Consistency of documentation, Follow-up recording of preventive interventions.

The lower compliance appears to be associated primarily with documentation inconsistency rather than complete absence of preventive action.

5.8 Key Patterns Identified

The analysis identified several important trends.

- **Strong Documentation Systems**: - The hospital maintains well-developed formal documentation processes.
- **Variation in Real-Time Compliance**: - Observed practice occasionally differed from recorded compliance.
- **Departmental Differences**: - Critical care areas demonstrated stronger adherence compared to general wards.
- **Impact of Workload**: - High patient volume influenced communication and documentation consistency.
- **Leadership Influence**: - Departments with closer supervision demonstrated better compliance.

5.9 Overall Findings

The study confirms that IPSG implementation has contributed significantly to strengthening patient safety practices within the hospital.

- **Major strengths include: -** Highly reliable patient identification systems, Strong medication management practices, Effective infection prevention measures.
- **Areas requiring improvement include: -** Communication consistency, Surgical team engagement, Fall prevention reassessment and documentation.

So, the results kind of suggest that patient safety turns out best when protocols are kept standardized , then monitored and reinforced too.

Meanwhile the staff interaction stuff, communication behavior, and this whole situational judgment side of it, seems to show more waver across different domains . In other words, it's less consistent

SUMMARY OF FINDINGS

The results of the study highlight the putting into practice of the International Patient Safety Goals by the Nurses and other various staffs. Found within tertiary care hospitals, safety goals appear useful for reinforcing patient protection measures. Their presence supports stronger systems without demanding attention. One sees improvement where these targets guide daily efforts. When applied carefully, outcomes often reflect better safeguards overall. Across the six IPSG areas, adherence reflects strong organizational dedication to upholding standards, yet cutting avoidable medical dangers remains a key aim.

Top results appeared in patient identification. Medication safety showed strong outcomes too. Infection prevention performed well, though slightly less consistent than the others. It appears that methods relying heavily on fixed guidelines fit more smoothly into everyday operations and clinical workflows. Effective staff training shows up in how tasks are handled daily. When monitoring systems work well, small issues get caught early. Administrative support strengthens these efforts behind the scenes.

On the practical ground, people moved in distinct ways; such variation highlights the value of ongoing observation. Staying alert matters more than a single test ever could. What counts is consistency, not isolated snapshots. Watching closely over time reveals what brief inspections miss entirely. Looking at everything together, results show putting IPSG into practice improved safety outcomes through fewer incidents. Errors in practice may drop when routines follow clearer guidelines. Still, uniform methods can support consistency across treatments.

CONCLUSION

The study links international patient safety goals to measurable improvements. Improving patient safety became more achievable through changes that reduced chances of mistakes during advanced medical treatment. A hospital appears in the record, yet feels slightly underdeveloped when reviewed directly within the document.

The findings generally indicate solid adherence across many IPSG areas, suggesting the hospital performs well and suggests that structured approaches shape how care quality improves when teams prioritize consistent

safety routines. The consistent compliance in patient identification, medication safety, and infection prevention demonstrates that standardized protocols , if correctly implemented and monitored, can greatly reduce preventable risks and improve reliability of clinical processes. The study also indicates that the IPSG is an effective framework for developing systematic approaches for patient safety. This framework aims to reduce variability in clinical practice, increase accountability and improve operational consistency across departments by standardizing key elements of healthcare delivery, in other words it makes things more steady and less ad hoc.

The very high compliance levels in the protocol areas show that a clearly defined procedure, together with regular monitoring and training of staff, can lead to major safety improvements.

The study also shows that organizational factors, such as leadership support, ongoing supervision, staff awareness, and workload management, are important in determining levels of compliance. Departments with more structured workflows and better supervision had higher levels of adherence than high-volume environments where operational pressures affected consistency. In summary, the study demonstrates that the use of IPSG is very effective in improving patient safety systems when integrated into routine healthcare procedures and not only used as an accreditation requirement. The findings provide useful evidence to healthcare administrators and quality managers that long-term improvement in patient safety requires continuous monitoring, institutional commitment and meaningful integration of safety standards within clinical behavior and organizational processes.

FUTURE SCOPE

The results of this study provide a solid foundation for further studies and quality improvement activities on patient safety and the implementation of the International Safety Goals. The present study gives important insights with regards to the levels of compliance in a tertiary care hospital however there is still a huge scope of extending this research to develop a larger evidence base and better patient safety strategies.

Later research could take place at multiple centers, spanning diverse medical contexts like state-run clinics, along with privately owned medical facilities, including specialized treatment hubs. Such studies might reveal clearer patterns in how IPSG guidelines are followed across various clinical settings. From differing settings, one might notice distinctions arising through structural choices alongside access to tools. These variations often reveal themselves when examining support systems across groups. Differences appear more clearly where frameworks shape availability. How things are set up influences what people can reach. Through such setups, gaps in opportunity emerge.

One way to look at hospital performance is through comparison, which might help set common standards across the country for patient care. Such comparisons could encourage facilities to align with national benchmarks over time. Looking closely at differences between institutions may lead to faster adoption of consistent practices. Insights from peer hospitals sometimes prompt updates in how services are delivered. When results are shared openly, it creates opportunities for gradual improvement in treatment approaches nationwide. Right now, the study works more like a snapshot - checking adherence just once, rather than tracking it across a longer time frame.

Later studies might look at how consistently IPSG rules are followed over extended periods, helping reveal whether results last. A clearer picture may emerge when patterns unfold across years instead of months. Tracking adherence closely could expose subtle shifts that shorter reviews miss entirely. Looking at results over time helps spot patterns. Through such analysis, the impact of safety measures might become clearer later on. Following these paths reveals how changes play out across years. With careful review, outcomes show what lasts. Observing sequences allows understanding of sustained effects carefully. Looking ahead, future efforts could shift into developing tailored training systems shaped around individual departments and approaches shaped by the specific shortcomings identified when reviewing adherence.

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